Bidirectional Mass Spectrometry Modeling Aided by Auxiliary Intermediate Steps From Graph Transformation Tools

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TBI Uni Vienna

2026-07-11

MS-Thesis

Bidirectional Mass Spectrometry Modeling Aided by Auxiliary Intermediate Steps From Graph Transformation Tools

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I will talk about my master's thesis project, which I am doing at the University of Vienna in the TBI group.

Outline

1. Problem
2. Background
3. Data
4. ML-Modelling
5. Results
6. Conclusions

Outline
Problem
Background
Data
ML-Modelling
Results
Conclusions

A quick roadmap: I start with the problem and some background on mass spectrometry, then the data and how I generate fragment candidates with MØD, then the machine-learning model and the results, and I close with the main takeaways.

What is my thesis about?

M . . . Molecular structure S . . . Mass spectrum

- Formulated mathematically as a function:

$$f : M \to S$$

- Investigation of the **inverse problem**:

$$f^{-1} : S \to M$$

- Integration of **chemically studied reaction rules** modeled in **MØD**
- Use of fragmentation trees to support traditional machine-learning-based MS models

What is my thesis about?
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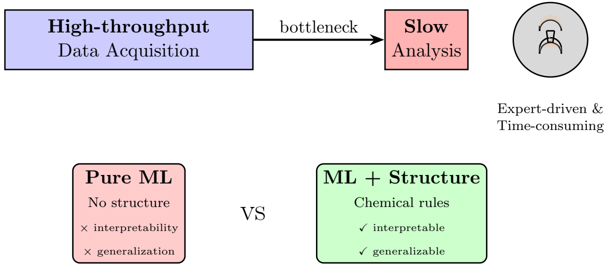
Integration of chemically studied reaction rules modeled in MØD
Use of fragmentation trees to support traditional machine-learning-based MS models

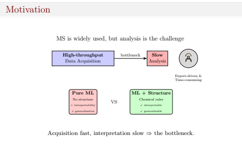
What is this project about?
I formulate MS modeling mathematically as a function mapping a molecular structure to its mass spectrum,
and I study the inverse problem of mapping a spectrum back to a structure.
The key idea is to bring in the relationships between the fragments, which I model in MØD, and use them to enhance the ML models.

Motivation

MS is widely used, but analysis is the challenge

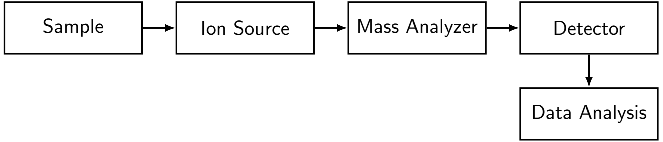


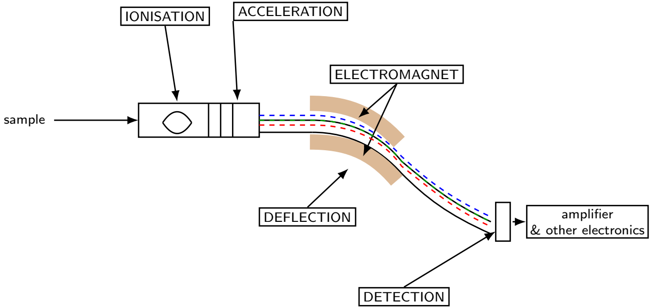
Acquisition fast, interpretation slow ⇒ the bottleneck.



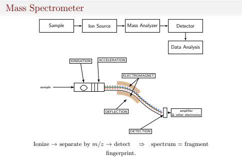
Why would we want to do MS modeling? What is the motivation?
Mass spectrometry is one of the most widely used tools for molecular analysis, but interpreting MS data is still largely expert-driven and time-consuming.
So the high throughput in data acquisition does not carry through to the analysis — that is the bottleneck.
And by incorporating chemical structure, we can make the ML models more interpretable and help them generalize.

Mass Spectrometer





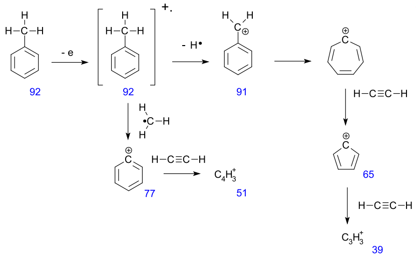
Ionize $\to$ separate by m/z $\to$ detect $\Rightarrow$ spectrum = fragment fingerprint.

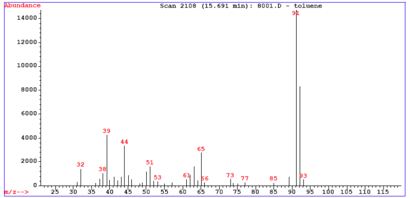


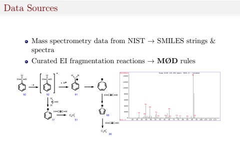
A quick reminder on how mass spectrometry works.
We have a sample, which is ionized;
the ions are then separated by their mass-to-charge ratio,
and detected to produce a spectrum.
Under electron ionization the molecule breaks into fragments, so the spectrum is essentially a fingerprint of those fragments — which is exactly what we exploit later.

Data Sources

- Mass spectrometry data from NIST → SMILES strings & spectra
- Curated EI fragmentation reactions → **MØD** rules

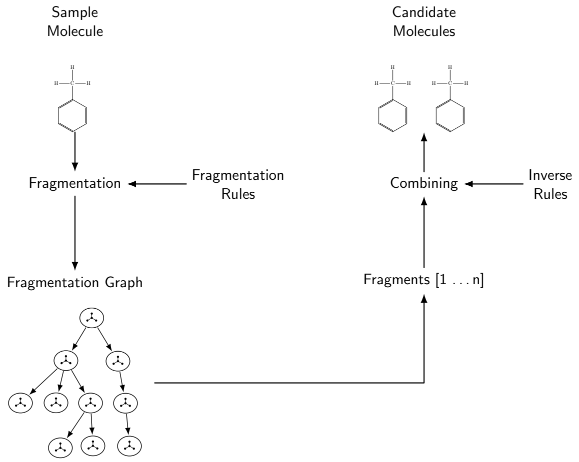


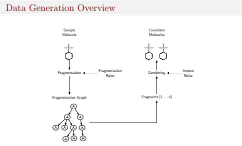




We have two main data sources:
Mass spectrometry data from NIST,
which gives us the molecular structure in the form of SMILES strings and spectra,
and curated EI fragmentation reactions, which are modeled as MØD rules.

Data Generation Overview





A sample molecule is fragmented using the fragmentation rules in MØD,
producing a fragmentation graph,
from which we extract the fragments.
These fragments are then combined using the inverse of the reaction rules
to produce candidate precursor structures for a given spectrum.

Sample Fragmentation Rules

$$\mathrm{R{-}CR_2{-}\overset{+\bullet}{Y}R} \xrightarrow{\alpha} \mathrm{R\bullet + CR_2{=}\overset{+}{Y}R} \tag{4.9}$$

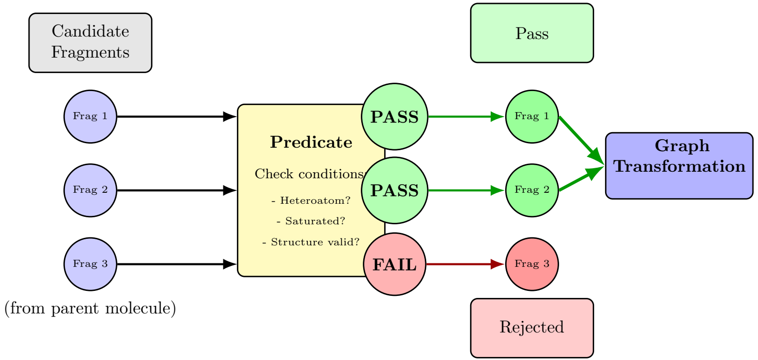
```
IMS_4_9 = mod.Rule.fromDFS(
    s =
    "[C]1[C]2[C]3([C]4)[_A+.]5"
    ">>"
    "[C.]1" "." "[C]2[C]3([C]4){=}[_A+.]5",
    name =
    "radical induced (alpha-)cleavage for a saturated site"
    " 4.9"
    " !R1R3R4Y5"
)
```

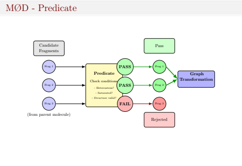

Sample Fragmentation Rules

At the top we see an example fragmentation rule as stated in the literature — note that it refers not only to individual atoms but also to whole groups (the R and Y placeholders).
At the bottom we see how the same rule is represented in code, using a notation that lets us encode those atom groups explicitly.

We use this extended notation in the graph-transformation step.

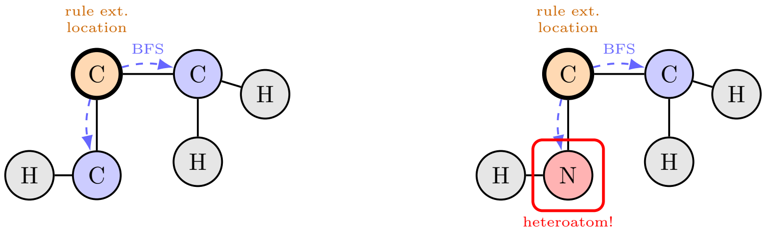
MØD - Predicate





In MØD, predicates act as callbacks during the transformation, which I use to filter candidates based on the rule extension.
In the next slides I will show examples of the predicates I implemented for this filtering.

MØD - Predicate - Alkyl Group

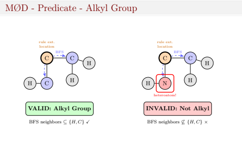


VALID: Alkyl Group

BFS neighbors $\subseteq \{H, C\}$ ✓

INVALID: Not Alkyl

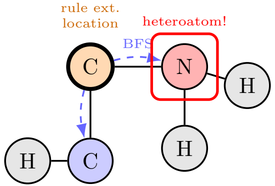
BFS neighbors $\not\subseteq \{H, C\}$ ×



A hydrocarbon (alkyl) substituent contains only C and H atoms. Starting from the rule-extension location, I collect neighbors in a breadth-first manner until I reach atoms that are already in the match of the original rule.

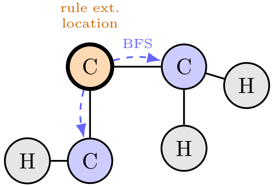
If every collected atom is a C or H, the group is a valid alkyl group; otherwise it is rejected.

MØD - Predicate - Heteroatom Group



VALID: Heteroatom Group

BFS $\setminus$ {H, C} $\neq \varnothing$ ✓



INVALID: No Heteroatom

BFS $\setminus$ {H, C} $= \varnothing$ ×

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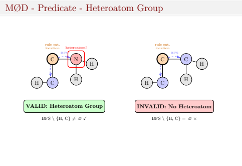
└Data
└MØD - Predicate - Heteroatom Group



A valid heteroatom group must contain at least one atom that is not C or H.
Again I collect the neighbors breadth-first, those not matched by the original rule,
and I check whether that set contains anything outside {H, C}.
If it does, the group is a valid heteroatom group; if it is a subset of {H, C}, it is rejected.

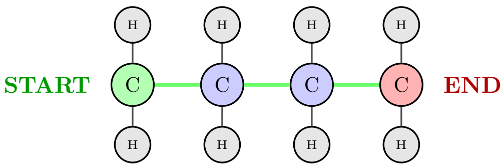
MØD - Predicate - Saturation

Goal: Find saturated path

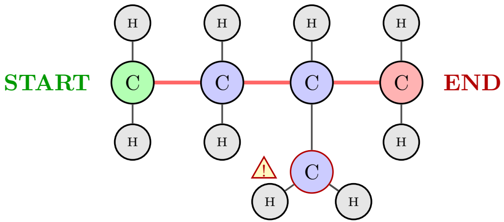
Requirements:
✓ Only carbon backbone
✓ Single bonds only (C—C)
✓ Only H side branches

Method:
Search from start to end following C-C single bonds

Saturated path



Non-Saturated path



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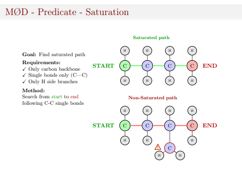
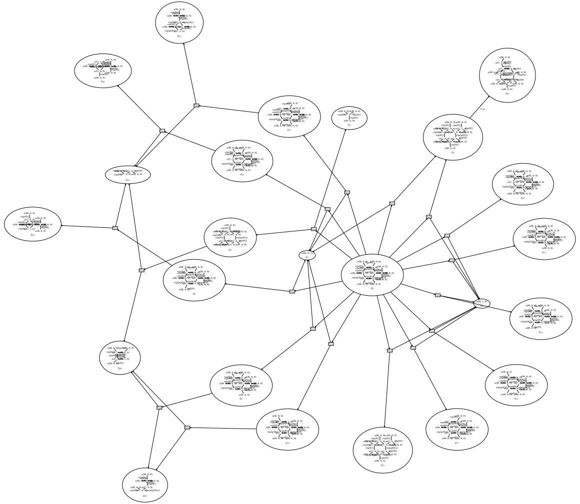
└Data
└MØD - Predicate - Saturation

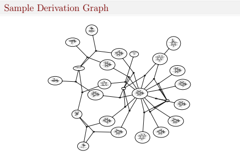


The saturation predicate checks for a chain of carbon atoms connected only by single bonds, where each carbon carries only hydrogen atoms as side branches.

I search from the start atom to the end atom along C–C single bonds: the top example is saturated, the bottom one fails because of the branching carbon and the double bond.

Sample Derivation Graph

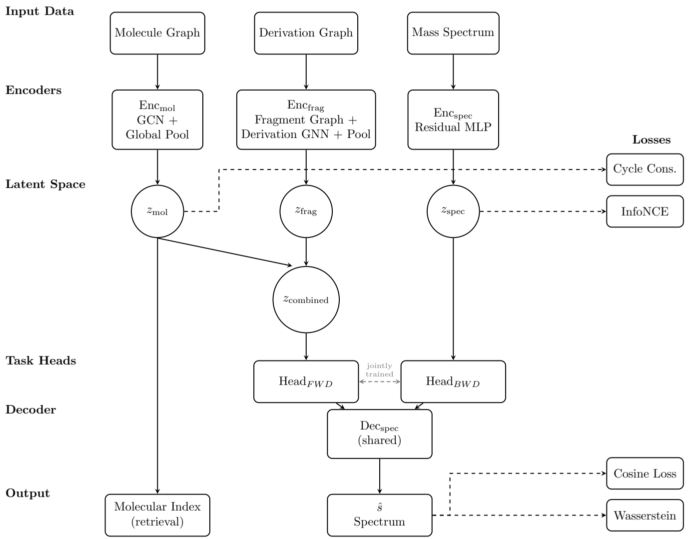




Here we see the computed output of the fragmentation, drawn as a derivation graph,
where the nodes are the fragments and the edges are the reactions that connect them.

The point is not to read every node — it is that from a single input molecule MØD generates a rich, connected space of fragments, and this is what we mine for candidate structures.

Architecture Diagram



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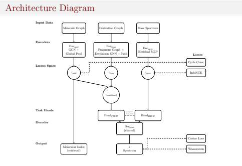
└ML-Modelling
└Architecture Diagram



This is the architecture of the model I implemented. It has three inputs: the molecule graph, the derivation (fragmentation) graph, and the mass spectrum.

Each input is mapped into one shared latent space by its own encoder — a GCN with global pooling for the molecule, a two-level graph encoder over the derivation graph for the fragments, and a residual MLP for the spectrum.

In the forward direction the molecular and fragment latents are fused and decoded into a predicted spectrum.

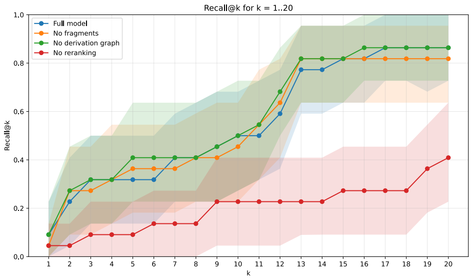
Importantly, the backward direction is not a spectrum-to-graph decoder: it is a retrieval path. The spectrum embedding is aligned with the molecular embeddings, so at test time I retrieve the nearest candidate structures by cosine similarity and then rerank them with the forward model.

The two directions share the task heads and decoder and are tied together by auxiliary losses — cycle-consistency and InfoNCE contrastive

Results – Bidirectional Retrieval

Proof-of-concept · ∼100 EI compounds · 22 test queries

Metric	Full model
Recall@1	0.091
Recall@5	0.318
Recall@10	0.500
Recall@20	0.864
MRR	0.243



⇒ candidate narrowing, not single-shot ID

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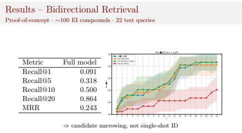
└Results
└Results – Bidirectional Retrieval



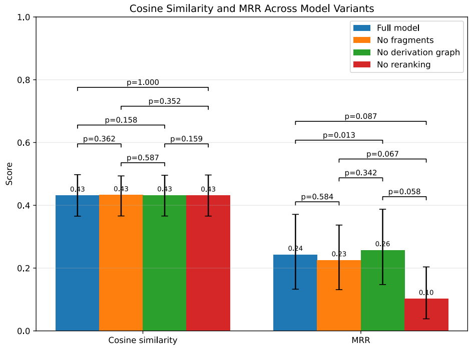
Now the results. This is a proof of concept on the order of a hundred EI compounds, with 22 held-out test queries.

Recall at k is the fraction of queries whose true structure is among the top-k retrieved candidates.

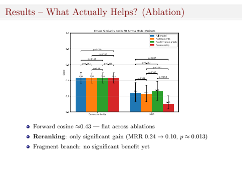
For the full model, Recall@1 is about 0.09, Recall@5 about 0.32, Recall@10 is 0.50, Recall@20 is 0.86, and the mean reciprocal rank is 0.24. So top-1 identification is limited, but the correct molecule is usually pulled into the top of the ranked list. The system behaves as a candidate-narrowing tool — which is still useful in a real structure-elucidation workflow — rather than a single-shot identifier.

Note the small test set: 22 queries means recall moves in steps of about 0.045, so I read differences between variants cautiously.

Results – What Actually Helps? (Ablation)



- Forward cosine ≈0.43 — flat across ablations
- **Reranking**: only significant gain (MRR 0.24 → 0.10, $p \approx 0.013$)
- Fragment branch: no significant benefit yet



To see what actually drives performance, I ran an ablation against the full model: no fragments, no derivation graph, and no reranking.
First, the forward task: all variants reach almost the same cosine similarity, around 0.43. Forward reconstruction is basically insensitive to the ablations.
Second, the one clearly significant effect is reranking. Removing it drops the mean reciprocal rank from 0.24 to 0.10, with a p-value around 0.013. So much of the retrieval benefit comes from combining latent-space retrieval with forward-model rescoring.
Third — and this is the honest part — the fragment and derivation-graph branches do not give a statistically significant gain yet. In its current form the fragment representation is largely redundant given the molecular encoder.

So the main contribution here is not a headline performance win from fragment augmentation; it is establishing a fragment-augmented bidirectional framework whose strengths and failure modes can be analysed

Conclusions

Contributions

- One bidirectional model: MØD graphs + two-level molecule + aligned spectral/structural embeddings ($f : M \to S$, $f^{-1} : S \to M$)
- Rule-extension DSL: alkyl / heteroatom / saturation
- Coupled objective: cosine + Wasserstein · InfoNCE · retrieval + rerank

Honest assessment

- Framework works; reranking helps
- Fragment branch: no significant gain yet
- Contribution = analysable framework, not accuracy win

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Contributions
One bidirectional model: MØD graphs + two-level molecule + aligned spectral/structural embeddings ($f : M \to S$, $f^{-1} : S \to M$)
Rule-extension DSL: alkyl / heteroatom / saturation
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Framework works; reranking helps
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Contribution = analysable framework, not accuracy win

To conclude, the contributions: to my knowledge this is the first framework to couple MØD-derived derivation graphs with a two-level molecular representation and to align spectral and structural embeddings in a single bidirectional model, covering both the forward and the inverse direction. Along the way I built a rule-extension DSL with the chemical predicates I showed, and a coupled forward–backward training objective.

And the honest assessment: the framework is functional and reranking clearly helps, but the fragment branch does not yet give a significant gain over the molecular encoder alone. So the value of this work is a bidirectional framework whose behaviour and failure modes I can analyse systematically — which sets up the outlook.

Outlook – Learn to *Generate* Fragmentation

Model *encodes* precomputed MØD graphs — it does *not* learn fragmentation.

Inspiration: **ICEBERG** [4] — *Generate – Score.*

- Have: the **Score** side
- Missing: a **Generate** module (which fragmentations?)
- MØD graphs → proxy supervision

prune/rank → autoregressive expansion → joint end-to-end

Outlook – Learn to *Generate* Fragmentation

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- Have: the **Score** side
- Missing: a **Generate** module (which fragmentations?)
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prune/rank → autoregressive expansion → joint end-to-end

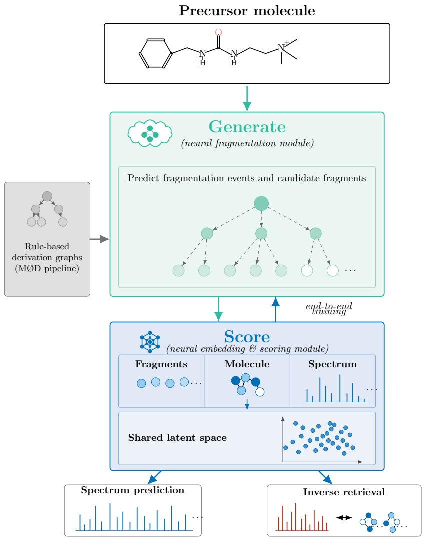
The clearest boundary of the current approach is that it does not learn fragmentation itself: it encodes derivation graphs that the rule-based MØD pipeline computes upstream. The learning problem only begins after the fragmentation space has been built.

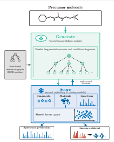
The natural next step, and my main direction for future work, is to move from encoding fragmentation to generating it. The key inspiration is ICEBERG, by Goldman and colleagues, which splits the task into a Generate module that proposes likely fragmentation events and a Score module that assigns intensities.

Seen this way, my model already supplies the Score side: it embeds molecules, fragments, and spectra into a shared space, predicts spectra, and aligns and retrieves. What is missing is the upstream Generate module that decides which fragments enter at all — and the symbolic MØD derivation graphs could serve as proxy supervision for training it.

I see a staged path: first learn to prune or rank derivation steps, then

Outlook – Toward a Generate–Score Architecture





This figure sketches that target architecture. A learned Generate module, trained with the MØD derivation graphs as proxy labels, proposes the fragmentation structure; the Score module — essentially my present model — embeds molecule, fragments, and spectrum in a shared latent space and produces both the forward spectrum prediction and the inverse retrieval. The two are trained end-to-end, so the model learns not only how to use fragments but which fragmentation structures to construct in the first place.

References

[1] V8rik. *Toluene Fragmentation.* 2008. URL: https://upload.wikimedia.org/wikipedia/commons/6/6c/TolueneFragmentation.svg (visited on 01/20/2026).

[2] Paginazero, Ronhjones. *Toluene Spectra.* 2017. URL: https://upload.wikimedia.org/wikipedia/commons/2/2c/Spettro_di_massa_del_toluene.PNG (visited on 01/20/2026).

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[4] Samuel Goldman, Janet Li, and Connor W. Coley. "Generating Molecular Fragmentation Graphs with Autoregressive Neural Networks". In: *Analytical Chemistry* 96.8 (Feb. 2024), pp. 3419–3428. ISSN: 1520-6882. DOI: 10.1021/acs.analchem.3c04654.

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[1] V8rik. Toluene Fragmentation. 2008. URL: https://upload.wikimedia.org/wikipedia/commons/6/6c/TolueneFragmentation.svg (visited on 01/20/2026).
[2] Paginazero, Ronhjones. Toluene Spectra. 2017. URL: https://upload.wikimedia.org/wikipedia/commons/2/2c/Spettro_di_massa_del_toluene.PNG (visited on 01/20/2026).
[3] Fred W McLafferty and František Tureček. Interpretation von Massenspektren. ger. Spektrum Lehrbuch. Berlin [u.a.]: Spektrum Akad. Verl., 1995. ISBN: 9783642398483.
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these are my sources and references

Discussion

Questions?

MS-Thesis
└─Conclusions
└─Discussion

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Discussion

Questions?

thank you for your attention
Do you have questions?